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Techno-economic analysis of off-grid hydrokinetic-based hybrid energy systems for onshore/remote area in South Africa

Kanzumba Kusakana

Department of Electrical, Electronic and Computer Engineering, Central University of Technology, Free State, Bloemfontein, South Africa

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ABSTRACT

Hydrokinetic power generation is a relatively recent type of hydropower that generates electricity from kinetic energy of flowing water making the conversion process more competitive compared to traditional micro-hydropower. Few authors have already analyzed the use of standalone hydrokinetic systems for rural electrification, however, there is no available literatures investigating the possibility of using this technology in combination with other renewable energy sources or diesel generator. Therefore, the aim of this paper is to investigate the potential use of hydrokinetic-based hybrid systems for low cost and sustainable electrical energy supply to isolated load in rural South Africa where adequate water resource is available. Different hybrid system configurations are modeled and simulated using the Hybrid Optimization Model for Electric Renewable (HOMER) and the results are analyzed to select the best supply option based on the net present cost and the cost of energy produced. The simulation results from two different case studies show that hybrid systems with hydrokinetic modules incorporated in their architectures have lower net present costs as well as lower costs of energy compared to all other supply options where the hydrokinetic modules are not included.

1. Introduction

The estimated electrification rate in South Africa is around 75% in 2013 [1]. There is still 25% of the country which is not electrified through the grid due to uneconomical cost of extension lines or difficult terrain especially in rural areas. These remote areas are generally electrified by means of standalone diesel generators (DG) which emit pollutants in the environment [2]. However, worldwide rise in fuel price as well as high transport and delivery cost to these isolated areas make the cost of energy produced from the DG very expensive [3].

South Africa has a potential of solar energy ranging from 4.5 to 6.5 kWh/m^2 and wind speed reaching annual averages of 6 m/s that can be used to generate sustainable amount of renewable energy [4,5]. These huge potential of renewable resources that can be used for energy generation is poorly used [6]. The installed capacity of solar, wind and hydropower plants in 2012 are approximately 30 MW, 10.1 MW and 3730.3 MW, respectively [7–9]; representing less than 2% of South African electricity production capacity.

The use of renewable energy source is still low in South Africa. The reason is due to the higher cost of energy produced by renewable sources compared to the one from the currently used coal power plants [10]. Referring to the Department of Mineral and Energy's aim (DME), setting a target of 10,000 GWh of renewable energy production by 2013 [11], it is imperative to explore the local renewable resources to provide sustainable electricity to the rural areas.

The solar photovoltaic (PV) is one of the renewable energy generation system mostly used in South Africa. The wind energy system is also being developed mostly in the coastal region where the wind potential is very good. However, for inland areas where adequate water resource is available, micro-hydro is the best supply option compared to other renewable resources in terms of cost of energy produced [12]. Unlike conventional hydropower technology, hydrokinetic is a relatively recent type of hydropower system that generates electricity by extracting kinetic energy of flowing water instead of potential energy of falling water, this makes hydrokinetic far less site specific and more competitive in terms of building cost compared to traditional micro-hydropower [13].

However the main disadvantage of hydrokinetic systems as well as of other renewable energy technologies is their resource-dependent output powers and their strong reliance on weather and climatic conditions [14]. Therefore, they cannot always match the fluctuating load energy requirements each and every time.

Using hydrokinetic-based hybrid systems for electricity generation can increase the reliability and reduce the cost of energy compared to the single energy systems. Hybrid systems are

E-mail address: kkusakana@cut.ac.za.

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well-suited solutions for supplying electricity to remote areas where the grid extension is not cost effective [15]; they can also decrease or eliminate the reliance on DGs in the remote areas [16].

Few feasibility studies have been conducted to develop stand-alone hydrokinetic power systems [17,18]; however currently there is no literature available showing the use of this technology operating in combination with other power generation systems [19]. Therefore, in this study the techno-economic analysis of hydrokinetic-based hybrid energy systems is done for two different loads located in two distinct South African remote areas and having different climatic conditions. HOMER (Hybrid Optimization Model for Electric Renewable) [20] is used as a simulation tool to build a computer model that optimizes the sizing and operation of different feasible hydrokinetic-based hybrid systems for the project's lifetime.

For this simulation, the main outputs to be analyzed are the hydrokinetic turbines (HKT), photovoltaic panels (PV), wind turbines (WT), DG and battery sizes (power ratings) with the intention of determining which hybrid system configuration or supply option is the best compared to the others based on the net present cost (NPC) and on the cost of energy produced (COE) selection criteria. For accurate results, all the resources, loads and system component data are actual data acquired in the specific studied locations.

2. Off-grid generation options

In this study, the supply options considered can be used alone or combined in hybrid systems configurations. Battery is incorporated as storage system and is also used to reduce the number of ON and OFF switching's of the backup DG in the hybrid systems simulated as suggested in Ref. [21]. The HKT, PV, WT and DG are the main supply options to be considered in this study which can be combined in deferent hybrid configurations. The energy model of each individual energy source is explained in the sections below. The emphasis will be on the hydrokinetic system which is a relatively new technology.

2.1. Hydrokinetic system

Hydrokinetic systems extract kinetic energy from moving water without the need for a dam, barrage or penstock. Hydrokinetic can generate power from low speed flowing water with almost zero environmental impact, over a much wider range of sites than those available for conventional hydropower generation [22].

In this study, the hydrokinetic system has been selected instead of the traditional micro-hydropower. Its operation principle is identical to the one of the wind turbine. Knowing that water is approximately 800 times denser than air [22], this means that the amount of energy produced by a hydrokinetic turbine is much greater than that produced by a wind turbine of equal diameter under equal water and wind speed. The other advantages of hydrokinetic system are that the water resource does not fluctuate unpredictably in a very short period of time as the wind speed, and the flow of water does not change in direction as the wind does.

The energy generated (E_{HKT}) by the hydrokinetic system is expressed as [23]:

$$E_{HKT} = \frac{1}{2} \times \rho_W \times A \times v^3 \times C_{p,H} \times \eta_{HKT} \times t \quad (1)$$

where ρ_W is the density of water (kg/m^3), $C_{p,H}$ is the coefficient of the hydrokinetic turbine performance, η_{HKT} is the combined efficiency of the hydrokinetic turbine and the generator, A is the turbine area (m^2), v is the water current velocity (m/s), and t is the time (s).

2.2. PV system

The sunlight striking on the panels, i.e. irradiance is measured in units of watts per square meter (W/m^2). The output energy (E_{PV}) of the solar PV system can be expressed as follows [24]:

$$E_{PV} = A \times \eta_m \times P_f \times \eta_{PC} \times I \quad (2)$$

where A is the total area of the photovoltaic generator (m^2), η_m is the module efficiency, P_f is the packing factor, η_{PC} is the power conditioning efficiency and I is the hourly irradiance (kWh/m^2).

2.3. Wind energy system

The energy generated (E_{WT}) by wind system is expressed as [25]:

$$E_{WT} = \frac{1}{2} \times \rho_a \times A \times v^3 \times C_{p,w} \times \eta_{WT} \times t \quad (3)$$

where ρ_a is the density of air (kg/m^3), A is the turbine area (m^2), v is the wind velocity (m/s), η_{WT} is the combined efficiency of the wind turbine and the generator, $C_{p,w}$ is the coefficient of the wind turbine performance, and t is the time (s).

2.4. Diesel generator

Diesel generators are normal diesel engines coupled to generator. To operate efficiently, most of DGs are designed in such a way that they always run between 80 and 100% of their kW rating while supplying the load.

The energy generated (E_{DG}) by a DG with rated power output (P_{DG}) is expressed as [26]:

$$E_{DG} = P_{DG} \times \eta_{DG} \times t \quad (4)$$

where η_{DG} is the efficiency of the DG.

3. Simulation data

Two case studies are conducted on two different sites from which the environmental data, load energy profile and system component costs are acquired and used as input to HOMER. Sections 3.1 and 3.2 describe the loads and the different renewable resources available on the two sites, respectively. Section 3.3 highlights the assumptions made while modeling the different supply options able to adequately respond to the loads energy requirements.

3.1. Case 1: rural household

3.1.1. Load description

A 24-h load data is obtained from a typical household situated in the Kwazulu Natal province at 30.6 degrees latitude south and 29.4 degrees longitude east and the corresponding load profile is illustrated on Fig. 1. The load is 5.6 kW peak and 35 kWh per day. It is assumed that any of the selected supply option should provide electricity for low consumption electrical appliances such as lights, TV, radio, laptop, fridge, kettle, cell phone chargers, iron, toaster, etc. [27]. When scrutinizing this load profile, one can notice a general pattern arising from the daily activities of the users which might change depending on different seasons of the year.

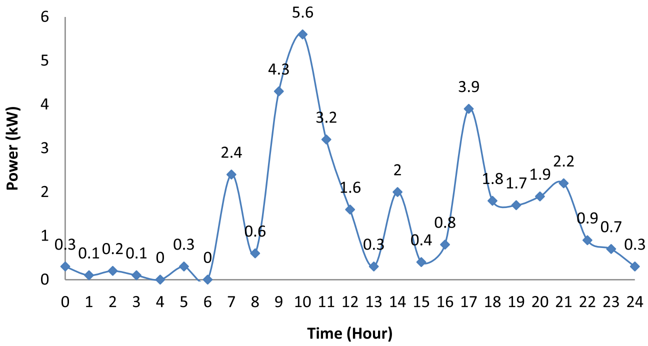

Fig. 1. Rural household load profile.

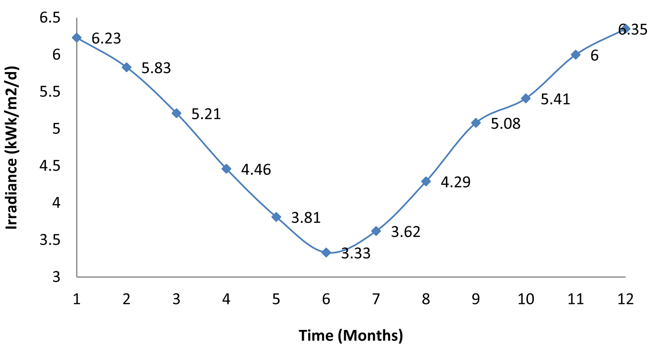

Fig. 2. Average daily radiation (case 1).

3.1.2. Renewable resources assessment

The solar and wind resources from the selected site are shown in Figs. 2 and 3, respectively [28]. One can notice that the water velocity and solar resources are good, but the wind resource is relatively low especially if it has to be used for micro-scale power generation. The average of the water velocity is given of Fig. 4 [17], it can be seen that the minimum water resource occurs during September; therefore the hydrokinetic plant has to be designed to generate the required electricity capacity during that month.

3.2. Case 2: base transceiver station (BTS)

3.2.1. Load description

The base transceiver station selected for this study is situated in the Western Cape region at 32.8 degrees latitude south and 17.9 degrees longitude east. The energy needed by the BTS communication equipment and the cooling system used to remove heat from

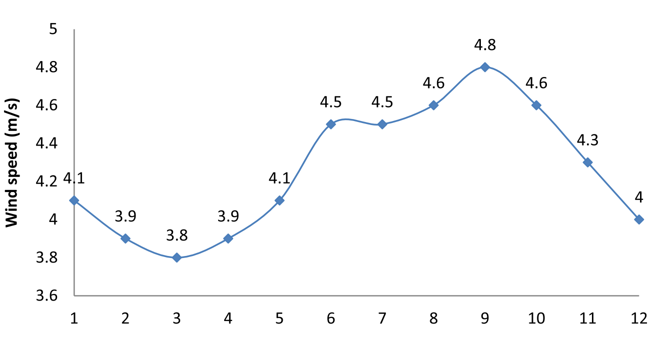

Fig. 3. Monthly average wind speed (case 1).

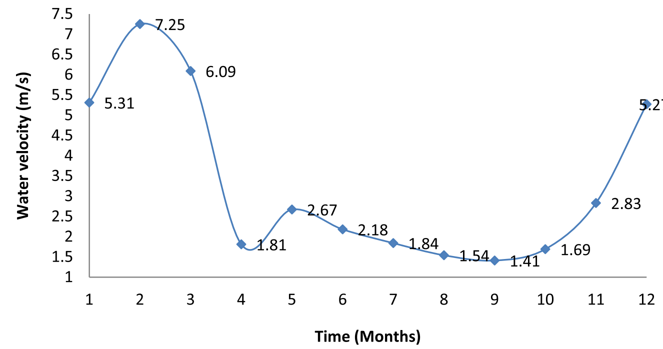

Fig. 4. Monthly average water velocity (case 1).

the cabin are given by Ref. [29]. The load profile resulting from the daily power demand of the radio, power conversion, antenna, transmission, security lights and cooling equipments at the base station site is given in Fig. 5. It is noticeable from this figure that except for the auxiliary equipments such as air-conditioning which is running during the day for only 6 h (11:00–17:00 h) and the security lights for 13 h throughout the night (18:00–7:00 h); the rest of the BTS communication equipment are running for 24 h non-stop giving a total peak demand of 3.8 kW and 59 kWh energy consumption per day.

3.2.2. Renewable resources assessment

The BTS site is situated at almost 1500 km from the selected rural household's site; therefore the two sites are in different climatic regions. The monthly average water velocity, solar irradiance and wind speed available on the BST site are illustrated in Figs. 6–8, respectively. As for case one, the hydrokinetic system designed to supply the BTS load must be able to operate during August which is the month with low hydro resources.

3.3. Components data and assumptions

HOMER does not incorporate hydrokinetic modules in its list of available components; therefore the hydrokinetic system has to be modeled as a wind turbine because of their similar operation principle (underwater wind turbine). A new wind turbine model can be created with the power curve of the hydrokinetic turbine showing power output versus water speed (which is entered as wind data). The anemometer height has to be set equal to the turbine hub height so that HOMER does not scale the wind speed data [30]. This implies that it is not possible to model or simulate a hybrid system having HKT and WT because HOMER can only take

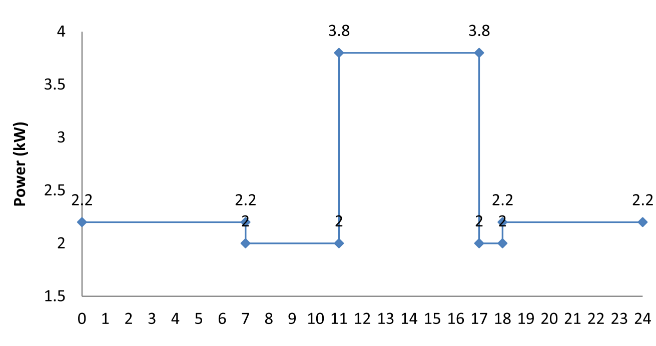

Fig. 5. BTS load profile.

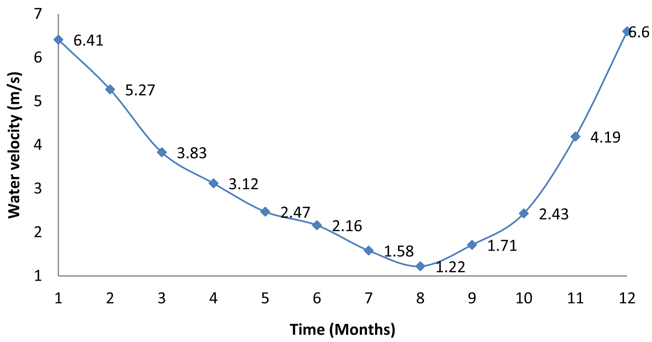

Fig. 6. Average water velocity (case 2).

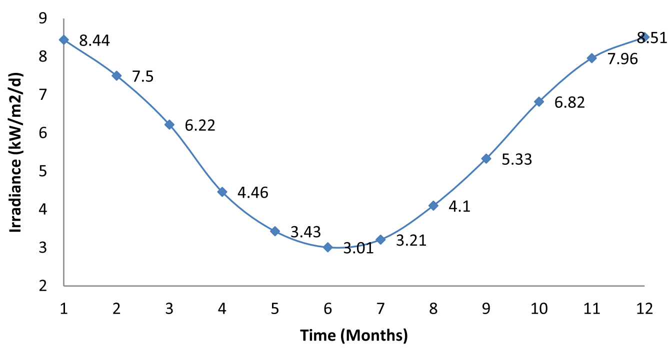

Fig. 7. Average daily radiation (case 2).

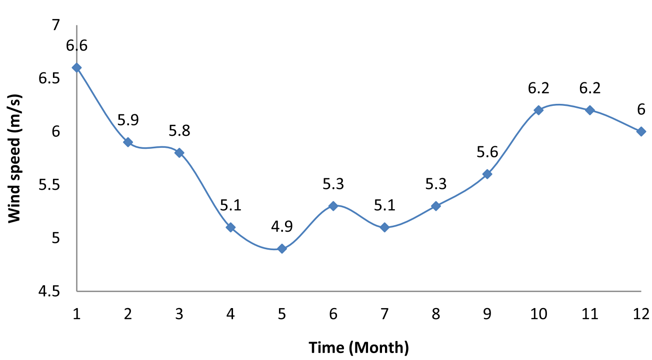

Fig. 8. Monthly average wind speed (case 2).

one set of wind resource at a time which can only be allocated to wind speed or water velocity not both at the same time.

The list of energy sources modules and their respective prices are presented in Table 1 which gives estimates based on quotations from local South African distributors of generation equipment [17,31,32].

Table 1
Component data and prices.

Modules	Price (US$)
HKT (1 kW)	7500
PV (1 kW)	3590
DG (7 kW)	1240
Converter (6 kW)	3730
Battery (6 V, 360 Ah)	215
WT (7.5 kW)	26,900

The followings are assumed for the components sizes and prices:

- The Darrieus hydrokinetic turbine (DHT) has been selected due to its ability of producing relatively high power output at low water speed [33]. This turbine has an area of 1.97 m^2 and a rated power of 1 kW at 1.4 m/s water velocity. Therefore it is assumed that there are no changes in the output power for water velocities above 1.5 m/s. Fig. 9 shows the corresponding turbine's power curve. The hydrokinetic turbine efficiency is assumed to be 65%, combined efficiency of the hydrokinetic turbine and the generator is assumed to be 60%, and the coefficient of power of the whole system is taken as 30%.
- The DHT operation and maintenance (O&M) cost is $20/year; and the system lifetime span is 25 years.
- The DG has an efficiency of 80% at full load and the price of diesel fuel and the lubricant are taken as 1.3$/l and 1.2$/l, respectively.
- The lifespan of the DG is 15,000 operating hours with an operation and maintenance of $0.50/kW.
- The O&M of the battery is taken to be $5/year.
- The converter lifespan is 15 years with an efficiency of 90% and the O&M cost of $10/kW.
- The lifespan of the PV panel is assumed to be 20 years with a derating factor of 90% and an O&M cost of $10/yr.
- The capacity shortage factor (CSF) which is the percentage of time when the supply option is not able to respond to the load energy requirements including the reserves, is set to 0% of the maximum hourly load.
- The cost of the grid extension is taken as $9000/km with an O&M cost of $180/km/yr. The cost of energy from the grid in South Africa is $0.144/kWh [34].

4. Evaluation criteria

The net present cost (NPC) of a system over its operation lifetime is used in this work as main criteria to find out the most feasible option. This indicator incorporates the initial costs (IC), O&M costs, replacement costs for each energy source in the architecture of the hybrid system, minus the salvage cost of each component [35].

The NPC ($) can be calculated using Eq. (5) below:

$$\mathrm{NPC} = \sum_j \left(I_j - S_j + \mathrm{OM}_j\right) \tag{5}$$

where j indicates the different component of a supply option; I_j, S_j, OM_j are the IC, present value of salvage and present value of operation and maintenance cost for component j, respectively.

The annualized cost of the system, which is the annualized value of the NPC, can be expressed as:

$$C_{\mathrm{an}} = \mathrm{CFR}\left(i, R_{\mathrm{proj}}\right) C_{\mathrm{NPC}} \tag{6}$$

where i is the annual real interest rate (%), R_{proj} is the operating lifespan of the supply option, and CRF is the capital recovery factor.

The cost of energy produced (COE) is the average cost per kWh of useful energy produced by the supply option. It can be expressed as:

$$\mathrm{COE} = \frac{C_{\mathrm{an}}}{E_{\mathrm{served}}} \tag{7}$$

where E_{served} is the total electrical load served (kWh/yr).

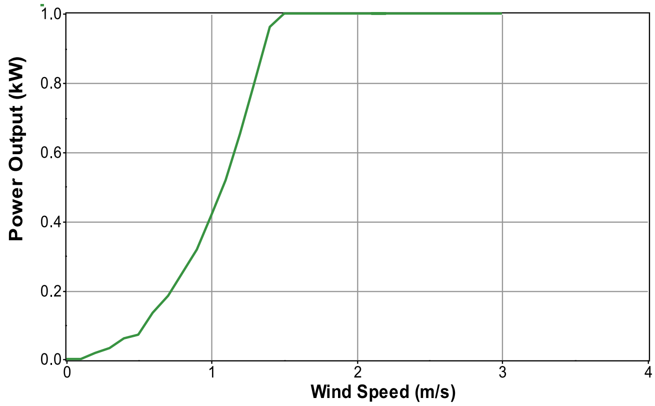

Fig. 9. Hydrokinetic power curve.

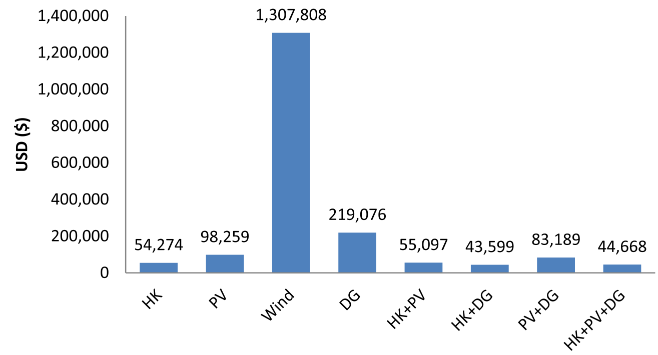

Fig. 10. Net present cost comparison of the different supply options (case 1).

The NPC and the COE as developed above will be used in this research work as main selection criteria of the different feasible supply options.

5. Results and discussion

HOMER simulates the different system configurations with all of the combinations of components that were given in the input section. It rejects from the potential results, all the combinations that are not feasible.

The simulation results obtained from the different supply options (as discussed in Section 2) will be analyzed and then compared based on their respective NPC and COE while supplying the same load.

5.1. Household case

Table 2 shows the optimal system architecture in terms of kW rating of the different supply options taken individually as well as in hybrid system configurations obtained after simulation. The number of batteries required for storage as well as the size of the converter is also indicated under the system architecture.

Figs. 10 and 11 compare the simulation results of the different options based on their NPC and COE. For the chosen site and load profile, the hybrid system (option 6) composed of HKT + DG, as described in Table 2, is the best of all the supply options under consideration.

Table 2
Architecture of different optimal supply options (household case).

Options	Description	Number of components and ratings (kW)
Option 1	HKT	HKT (4 kW)/18 Batteries/Converter (7 kW)
Option 2	PV	PV (15 kW)/68 Batteries/Converter (7 kW)
Option 3	WT	30 WT (7.5 kW)/919 Batteries/Converter (7 kW)
Option 4	DG	DG (7 kW)
Option 5	HKT + PV	HKT (3 kW)/PV(2 kW)/24 Batteries/Converter (7 kW)
Option 6	HKT + DG	HKT (3 kW)/DG (1 kW)/12 Batteries/Converter (7 kW)
Option 7	PV + DG	PV (9 kW)/DG (2 kW)/48 Batteries/Converter (7 kW)
Option 8	HKT + PV + DG	HKT (3 kW)/PV(1 kW)/DG (1 kW)/13 Batteries/Converter (7 kW)

5.1.1. Summary of the possible supply options

Table 3 gives a summary of the technical, economical as well as environmental results of the simulated supply options. It can be noticed that for the site considered, the WT system has the highest NPC as well as COE which makes it uneconomical to supply the selected load. Therefore hybrid systems including WT in their architecture will not be considered.

From Table 3, we can notice that based on the NPC, COE and the breakeven grid extension distance that the system composed of HKT + DG (option 6) is the best supply option for the load under consideration. The use of option 6 can significantly reduce the amount of fuel consumed as well as the pollutant emitted by 97% compared to the option 4 where the DG is used alone to supply the load. This reduction corresponds to almost US$ 9298 annual savings. However, if the environmental concern (zero pollutant emitted) is added as one of the main selection criteria, then option 1 (HKT-alone) can be selected even though its NPC and COE are higher than option 6 as described in Table 3.

5.1.2. Further analysis of the best supply option: HKT + DG (option 6)

The hybrid system HKT + DG (option 6) has IC of $28,988; an NPC of $43,599, an OC of $1143/yr and a COE of 0.265$/kWh.

Figs. 12–14, respectively, show the monthly average electric contribution of the hybrid system and both the HKT and the DG taken individually. It can be seen that the HKT system is able to provide enough energy to supply the load most of the time. However, the following operation cases can be observed:

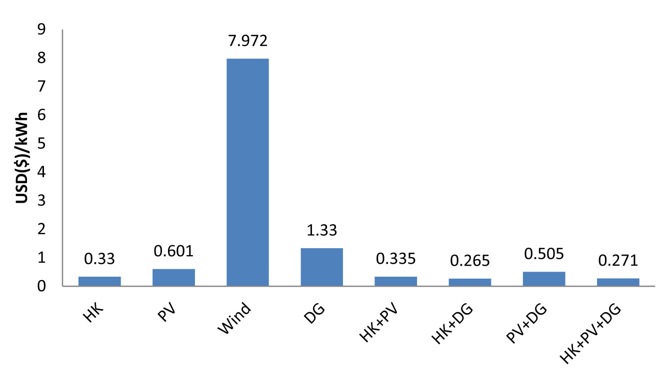

Fig. 11. Cost of energy produced comparison of the different supply options (case 1).

Table 3
Summary of the different supply options simulation results (household case).

Item	HKT	PV	Wind	DG	HKT + PV	HKT + DG	PV + DG	HKT + PV + DG
Capital ($)	37,601	72,201	1,008,316	1,270	38,571	*28,988*	46,715	32,793
NPC ($)	54,274	98,259	1,307,808	219,076	55,097	***43,599***	83,189	44,668
COE ($/kWh)	0.33	0.601	7.972	1.33	0.335	***0.265***	0.505	0.271
Operation ($/year)	1304	2038	23,428	17,041	1288	*1143*	2853	929
Replacement ($)	17,442	31,070	313,808	8164	17,674	*8939*	25,869	8170
Fuel (L/year)	0	0	0	9539	0	*241*	0	147
Salvage ($)	−3071	−11,403	−101,945	−178	−3893	*−827*	−7295	−1092
Grid extension (km)	2.7	6.6	114	17.3	2.77	*1.76*	5.26	1.85
Carbon dioxide (kg/year)	0	0	0	25,120	0	*634*	1816	386

The bold-italicized values are the cost of energy produced as well as the net present cost (selection criteria) of the selected supply option.

- During April: Due to the decrease of water resource, the hydrokinetic output power is reduced. Therefore more energy is drawn from the battery to supply the load during peak times.
- From August to October: There is insufficient water resource and the HKT used with the battery storage system cannot provide enough energy to totally respond to the load requirements. Therefore the DG is used in conjunction with the HKT and the battery storage system to bring the balance of energy needed by the load.

Figs. 15 and 16 show the inverter output power and the battery state of charging, respectively. It can be seen that throughout the

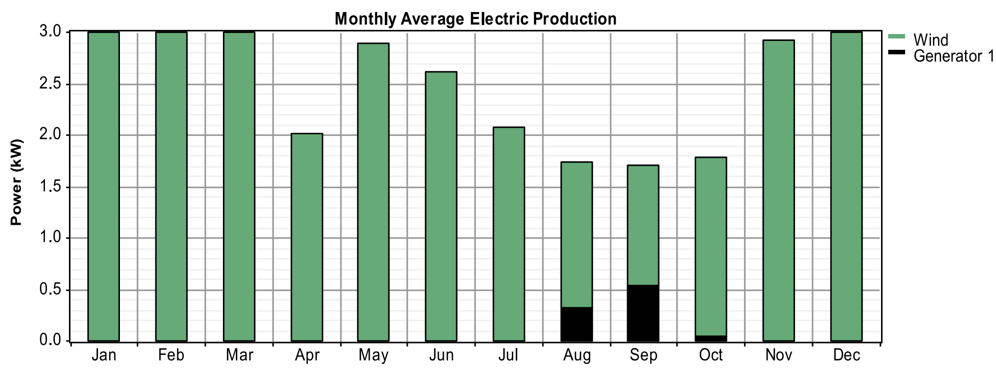

Fig. 12. Hybrid system monthly average electric production (case 1).

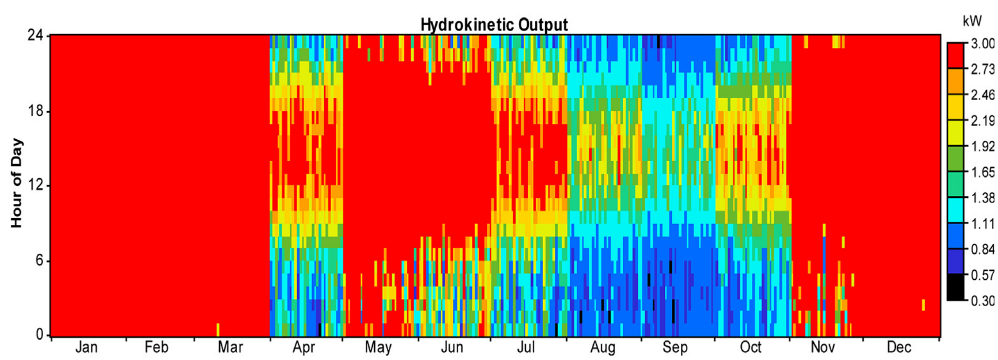

Fig. 13. Hydrokinetic monthly electric production (case 1).

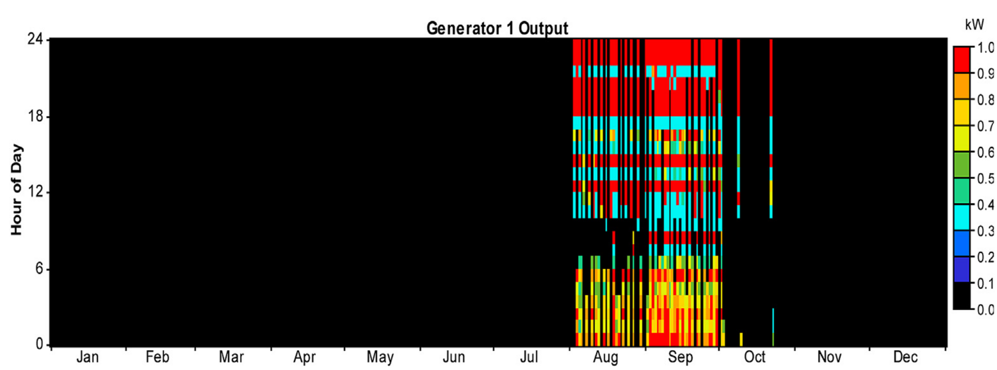

Fig. 14. Diesel generator electric production (case 1).

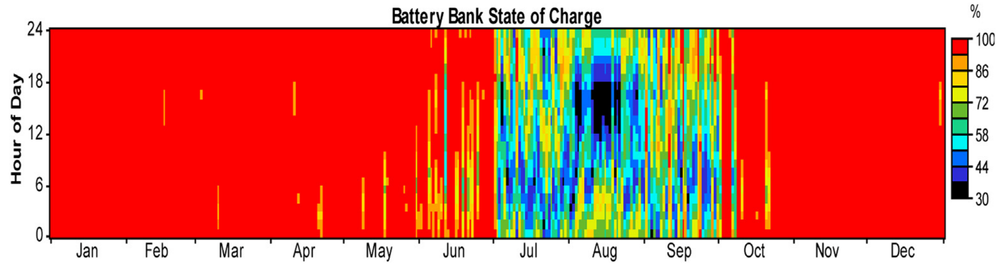

Fig. 15. Battery state of charge (case 1).

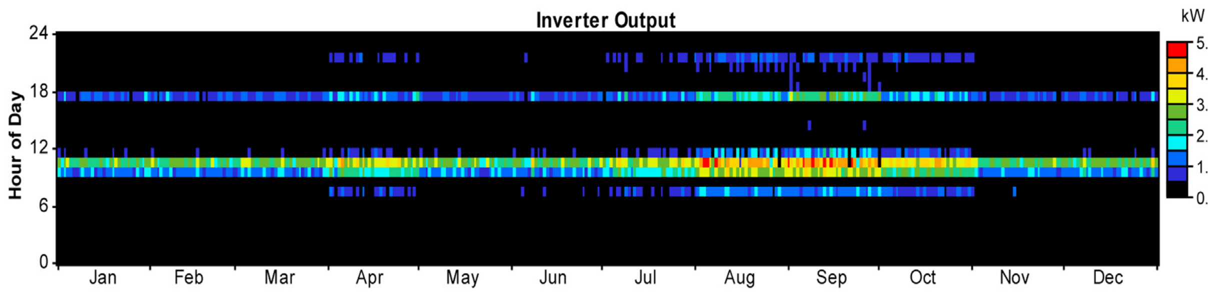

Fig. 16. Inverter output power (case 1).

year the battery system is charged during off-peak times and used during peak power demands occurring before noon (09:00 h–12:00 h) as well as in the evening times around 5pm. But in April and from August to October, the load is also heavily relying on the power stored in the batteries most of the day to compensate the power deficit from HKT system. Therefore the battery state of charge during these 4 months is lower compared to the other months of the year.

Fig. 17 gives the breakeven grid extension distance at 1.76 km. This means that the total hybrid system project cost over 25 years will be equivalent to building an extension line of 1.76 km.

5.2. *Case 2: base transceiver station (BTS)*

As for the case 1, the optimal sizes of the different supply options able to supply the BTS load under the different technical, economical and environmental constraints are presented in Table 4.

Figs. 18 and 19 present the comparison of the simulation results of the different options based on the NPC and COE. For the BTS load, the hybrid system (option 6) composed of HKT + DG, as described in Table 4, is the best of all the supply options under consideration.

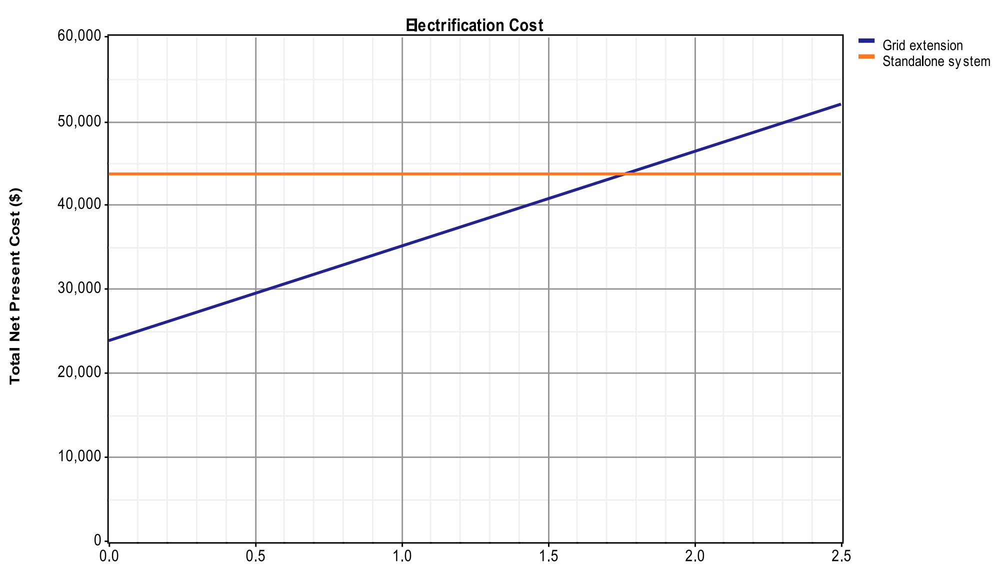

Fig. 17. Grid extension distance (case 1).

Table 4
Architecture of different optimal supply options (BTS case).

Options	Description	Number of components and ratings (kW)
Option 1	HKT	HKT (9 kW)/5 Batteries/Converter (5 kW)
Option 2	PV	PV (32 kW)/107 Batteries/Converter (5 kW)
Option 3	WT	73 WT (7.5 kW)/466 Batteries/Converter (5 kW)
Option 4	DG	DG (5 kW)
Option 5	HKT + PV	HKT (8 kW)/PV (1 kW)/8 Batteries/Converter (5 kW)
Option 6	HKT + DG	HKT (4 kW)/DG (2 kW)/8 Batteries/Converter (5 kW)
Option 7	PV + DG	PV (7 kW)/DG (3 kW)/2 Batteries/Converter (5 kW)
Option 8	HKT + PV + DG	HKT (4 kW)/PV (1 kW)/DG (2 kW)/7 Batteries/Converter (5 kW)

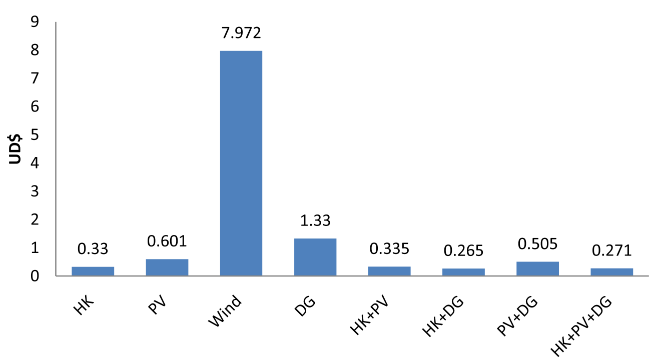

Fig. 18. Net present cost comparison of the different supply options (case 2).

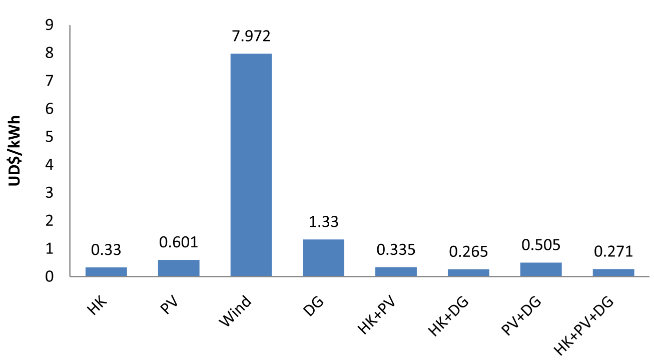

Fig. 19. Cost of energy produced comparison of the different supply options (case 2).

5.2.1. Summary of the possible supply options

Table 5 gives a summary of the technical, economical as well as environmental results of the different simulated supply options. As for case 1, the WT will not be considered in hybrid systems configuration because of the non-economical characteristic of its NPC and COE.

From this table we can notice that based on the NPC, COE and the breakeven grid extension distance that the system 6 (HKT + DG) is the best of all the options considered in the simulation. The use of option 6 can significantly reduce the amount of fuel consumed as well as the pollutant emitted by 93% compared to the option 4 where the DG is used alone to supply the load. This reduction corresponds to almost US$ 10,725 annual savings. However, if the environmental concern (zero pollutant emitted) is added as one of the main selection criteria, then option 5 (HKT + PV) can be selected even though its NPC and COE are higher than option 6 as described in Table 5.

5.2.2. Further analysis of the best supply option : HKT + DG (option 6)

Option 6 (HKT + DG) has IC of $35,183; an NPC of $51,887, an OC of $1307/yr and a COE of 0.189$/kWh. Figs. 20–22, respectively, show the monthly average electric contribution of the hybrid system and both the HKT and the DG taken individually. It can be seen that the HKT system is able to provide enough energy to supply the load most of the time. However, from July to September there is insufficient water resource and the HKT used with the storage system cannot provide enough energy to totally respond to the load requirements. Therefore the DG is used in conjunction with the HKT and the battery storage system to bring the balance in energy needed by the load especially during the time of the day when the air-conditioning system is ON.

Figs. 23 and 24 show the inverter output power and the battery state of charging, respectively. It can be seen that from July to September, the load is also heavily relying on the power stored in the batteries especially from 11h:00 to 17h:00 when the air-conditioning system requires more energy from the hybrid system. Therefore the battery state of charge during these 3 months is lower compared to the other months of the year.

Fig. 25 gives the breakeven grid extension distance at 1.1 km, meaning that the total hybrid system project cost over 25 years will be equivalent to building an extension line of 1.76 km.

6. Conclusion

This paper investigated the use of hydrokinetic-based hybrid systems to supply electricity to isolated areas of South Africa where adequate water resource is available. From the review of the few literatures available on hydrokinetic, it has been revealed that no

Table 5
Summary of the different supply options simulation results (BTS case).

Item	HKT	PV	Wind	DG	HKT + PV	HKT + DG	PV + DG	HKT + PV + DG
Capital ($)	71,684	140,994	650,799	886	68,419	*35,183*	29,201	38,558
NPC ($)	76,277	186,170	961,854	194,547	74,054	***51,887***	135,803	54,175
COE ($/kWh)	0.278	0.681	3.520	0.709	0.270	***0.189***	0.495	0.197
Operation ($/year)	359	3534	24,333	15,149	441	*1307*	8339	1222
Replacement ($)	2233	57,136	316,936	6353	3913	*3139*	12,858	4045
Fuel (L/year)	0	0	0	8870	0	*619*	4788	540
Salvage ($)	−367	−22,997	−54,436	−83	−1069	*−285*	−4739	−913
Grid extension (km)	3.25	13	81.6	13.7	3.06	*1.1*	8.52	1.3
Carbon dioxide (kg/year)	0	0	0	23,356	0	*1631*	12,608	1421

The bold-italicized values are the cost of energy produced as well as the net present cost (selection criteria) of the selected supply option.

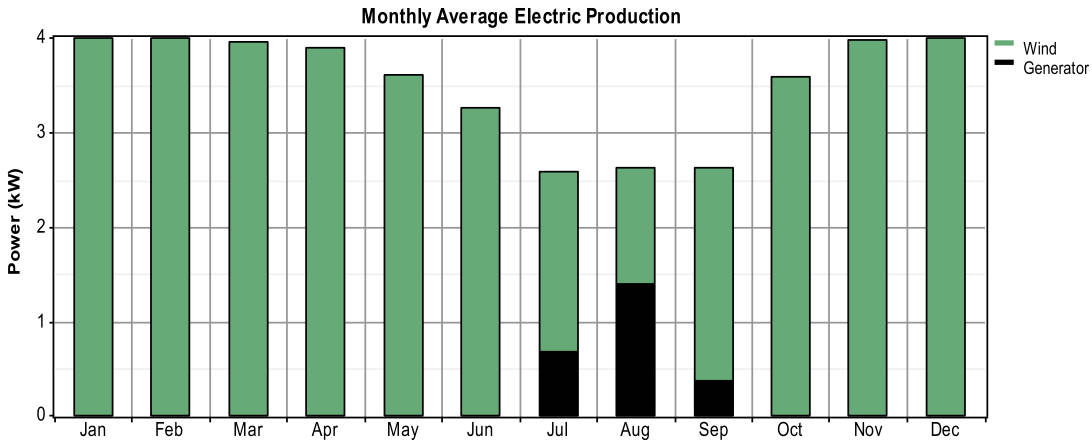

Fig. 20. Hybrid system monthly average electric production (case 2).

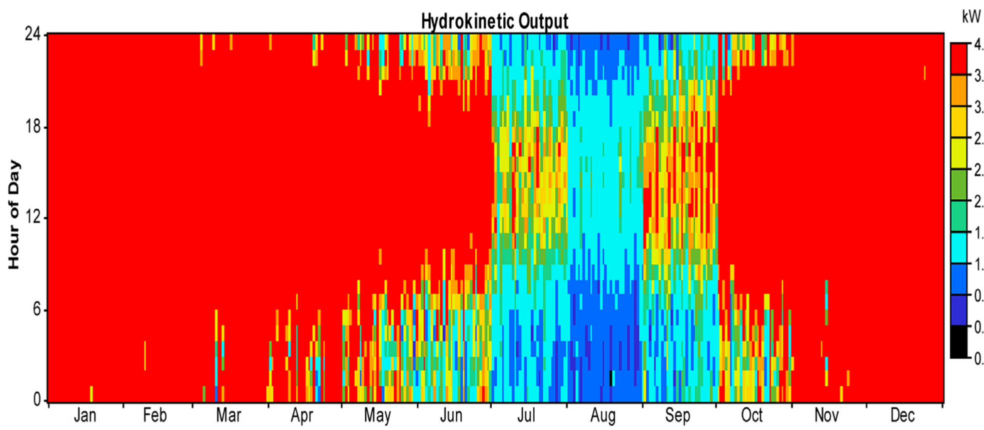

Fig. 21. Hydrokinetic monthly electric production (case 2).

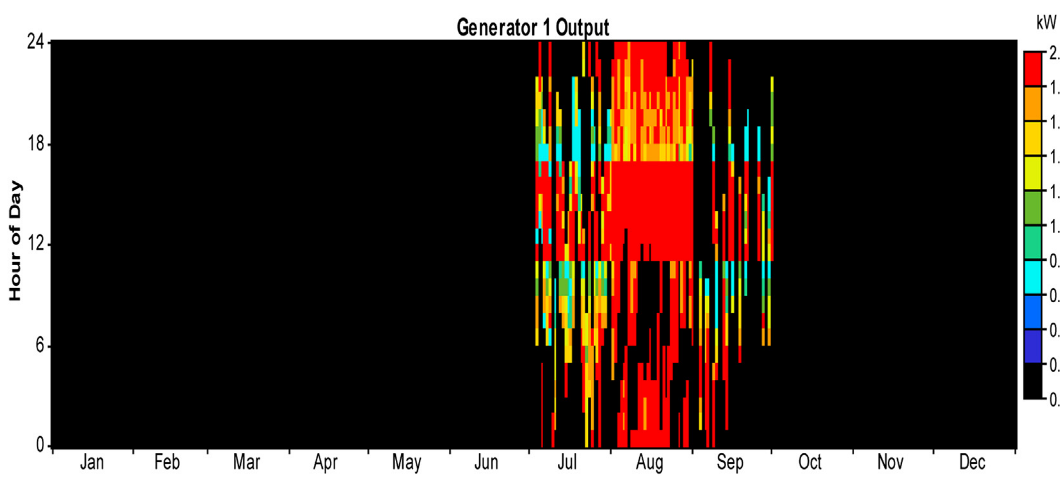

Fig. 22. Diesel generator electric production (case 2).

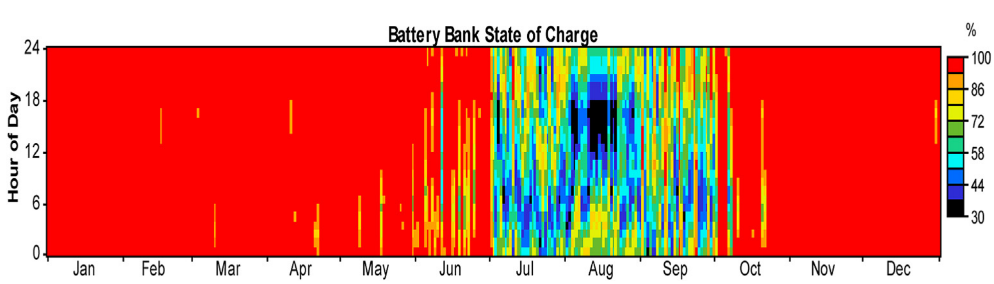

Fig. 23. Battery state of charge (case 2).

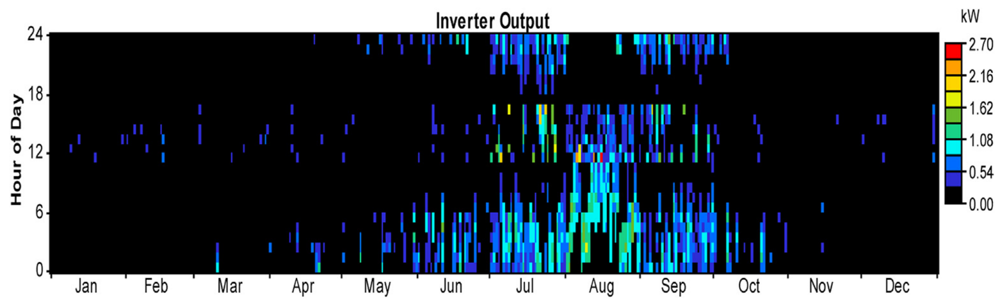

Fig. 24. Inverter output power (case 2).

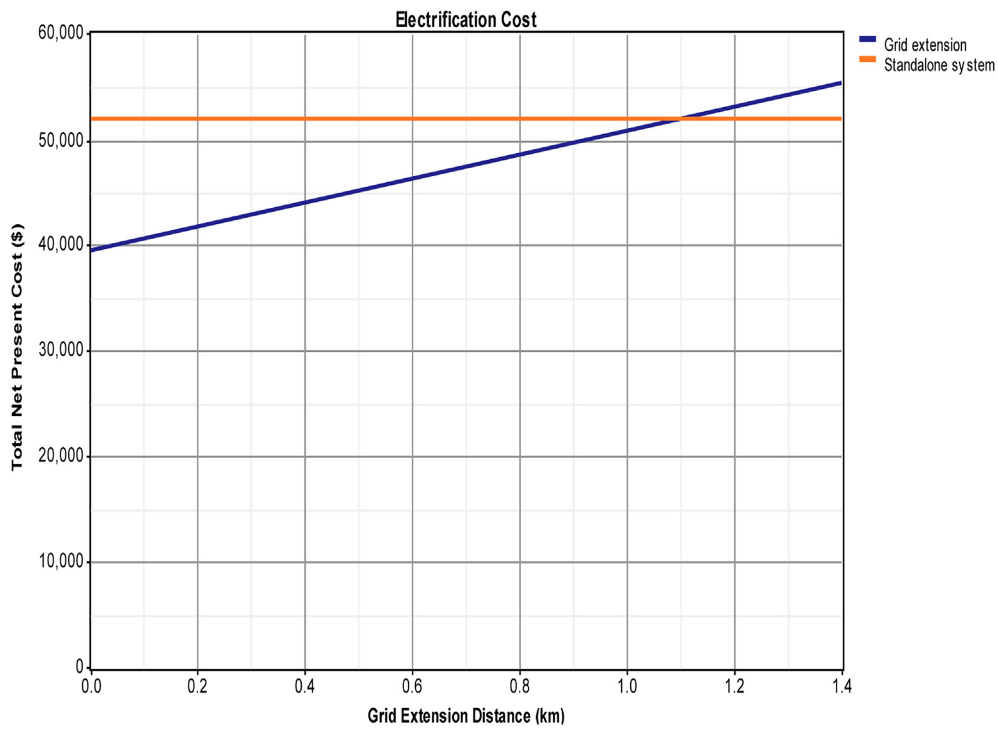

Fig. 25. Grid extension distance (case 2).

work have been published up to so far dealing with the use of this technology operating in hybrid combination with other energy sources such as PV, wind or diesel generator. A typical rural household and a base transceiver station located in different climatic regions and having different load patterns have been used as case studies for techno-economic analysis with the aim of finding the best supply option.

Simulations of eight different options (HKT, PV, WT, DG, HKT + PV, HKT + DG, PV + DG and HKT + PV + DG) have been performed with HOMER. The results have been analyzed and the best supply option has been selected based on the net present cost and the cost of energy produced with 0% capacity shortage as constraint.

For the case of the rural household, the hybrid system composed of HKT (3 kW)/ DG (1 kW)/12 Batteries/Converter (7 kW) is the best supply option. It has an initial cost of $28,988, a net present cost of $43,599 and a cost of energy produced of $0.265/kWh. For the second case where the base transceiver station is involved, the hybrid system composed of HKT (4 kW)/DG (2 kW)/8 Batteries/ Converter (5 kW) is the best supply option. It has an initial cost of $35,183, a net present cost of $51,887 and a cost of energy produced of $0.189/kWh.

For both case studies, the simulation results have also revealed that hybrid systems having hydrokinetic modules in their architectures have lower net present costs as well as lower cost of energy compared to all other supply options where the hydrokinetic modules are not included. Apart for sensibly decreasing the costs, hydrokinetic modules heavily contribute to the reduction of pollutant emitted by diesel generators when combined in hybrid systems configurations.

It has been noticed that HOMER cannot simulate hybrid systems which incorporate wind and hydrokinetic modules at the same time. Thus, for further studies, it would be interesting to develop proper mathematical models for combined optimal sizing and operation control of hydrokinetic-based hybrid systems allowing the impact analysis of components such as wind modules on the net present cost or cost of energy produced.

References

[1] van Ruijven BJ, Schers J, van Vuuren DP. Model-based scenarios for rural electrification in developing countries. Energy 2012;38:386–97.

[2] Kusakana K, Vermaak HJ. Hybrid diesel generator – battery systems for off-grid rural applications. In: IEEE international conference on industrial technology (ICIT 2013), 25–28 February 2013, Cape Town; 2013. pp. 839–44.

[3] Mahmoud MM, Ibrik IH. Techno-economic feasibility of energy supply of remote villages in Palestine by PV-systems, diesel generators and electric grid. Renewable and Sustainable Energy Reviews 2006;10:128–38.
[4] Munzhedzi R, Sebitosi AB. Redrawing the solar map of South Africa for photovoltaic applications. Renewable Energy 2009;34:165–9.
[5] van der Linde HA, Sayigh AAM. The economics of wind energy in South Africa. Renewable Energy 1999;16:869–71.
[6] Pegels A. Renewable energy in South Africa: potentials, barriers and options for support. Energy Policy 2010;38:4945–54.
[7] Maphelele T, Stanford R, Kooverji B. South Africa solar energy technology road map: solar PV baseline report (draft 2). Available from: http://www.sapvia.co.za/wp-content/uploads/2013/03/Solar-PV-Baseline-for-South-Africa_Draft-2_2013-06-26.pdf; May 2013 [accessed 28.12.13].
[8] Hagemann K. South Africa's REIPPPP: a blueprint for wind power in Africa?. 2013 Africa energy yearbook. Available from: http://www.energynet.co.uk/webfm_send/286; 2013 [accessed 28.12.13].
[9] Klunne WJ. Current status and future developments of small and micro hydro in southern Africa. In: Hidroenergia 2012, Wrocław, Poland; 23–26 May 2012.
[10] de Groot RAF, van der Veen VG, Sebitosi AB. Comparing solar PV (photovoltaic) with coal-fired electricity production in the centralized network of South Africa. Energy 2013;55:823–37.
[11] DME, Department of Minerals and Energy Affairs. see SOUTH AFRICA "White paper on renewable energy"; November 2003.
[12] Paish O. Small hydropower: technology and current status". Renewable and Sustainable Energy Reviews 2002;6:537–56.
[13] Vermaak HJ, Kusakana K, Koko SP. "Status of micro-hydrokinetic river technology in rural applications: a review of literature. Renewable and Sustainable Energy Reviews 2014;29:625–33.
[14] Chen F, Duic N, Alves LM, Carvalho MG. Renewislands – renewable energy solutions for islands. Renewable and Sustainable Energy Reviews 2007;11(8):1888–902.
[15] Kusakana K, Vermaak HJ. Hybrid renewable power systems for mobile telephony base stations in developing countries. Renewable Energy 2013;51:419–25.
[16] Dekker J, Nthontho M, Chowdhury S, Chowdhury SP. Economic analysis of PV/diesel hybrid power systems in different climatic zones of South Africa. Electrical Power and Energy Systems 2012;40:104–12.
[17] Kusakana K, Vermaak H. Feasibility study of hydrokinetic power for energy access in rural South Africa. In: Proceedings of the IASTED Asian conference, power and energy systems; 2–4 April 2012. p. 433–8.
[18] Kusakana K, Vermaak HJ. Hydrokinetic power generation for rural electricity supply: case of South Africa. Renewable Energy 2013;55:467–73.
[19] Alaska Hydrokinetic Research Center (AHERC). Draft strategic plan 2010–2013. Available from: http://www.uaf.edu/files/acep/2010_10_24_AHERC_Strategic_Plan.pdf [accessed 13.12.13].
[20] National Renewable Energy Laboratory (NERL). Hybrid optimization model for electric renewable (HOMER). Available from: www.nrel.gov; 2005 [accessed 17.12.13].
[21] Tazvinga H, Xia X, Zhang J. Minimum cost solution of photovoltaic–diesel–battery hybrid power systems for remote consumers. Solar Energy 2013;96:292–9.
[22] Güney MS, Kaygusuz K. Hydrokinetic energy conversion systems: a technology status review. Renewable and Sustainable Energy Reviews 2010;14(9):2996–3004.
[23] Maniaci DC, Ye Li. Investigating the influence of the added mass effect to marine hydrokinetic horizontal-axis turbines using a general dynamic wake wind turbine code. In: Oceans' 11 conference, 19–21 September 2011, Hawaii, USA. Clark RH. Elements of tidal-electric engineering. Wiley-IEEE Press; 2007.
[24] Davila-Vilchis JM, Mishra RS. Performance of a hydrokinetic energy system using an axial-flux permanent magnet generator. Energy 2014;65:631–8.
[25] Singh GK. Solar power generation by PV (photovoltaic) technology: a review. Energy 2013;53:1–13.
[26] Elizondo J, Martinez J, Probst O. Experimental study of a small wind turbine for low- and medium-wind regimes. International Journal of Energy Research 2009;33:309–26.
[27] Hu Y, Solana P. Optimization of a hybrid diesel-wind generation plant with operational options. Renewable Energy 2013;51:364–72.
[28] Dekker J, Nthontho M, Chowdhury S, Chowdhury SP. Investigating the effects of solar modelling using different solar irradiation data sets and sources within South Africa. Solar Energy 2012;86:2354–65.
[29] Clean Energy Decision Support Center. RETScreen software. Available from: www.retscreen.net [accessed 17.10.13].
[30] Anayochukwu AV. Optimal control of PV/wind/hydro-diesel hybrid power generation system for off-grid macro base transmitter station site. Electronic Journal of Energy & Environment August 2013;1(2).
[31] HOMER support "Modeling a hydrokinetic turbine in HOMER". Available from: http://support.homerenergy.com/index.php?/Knowledgebase/Article/View/136/50; 2013 [accessed 27.11.13].
[32] Component price list acquired from MLT drives. Available from: www.mltdrives.com; 2013 [accessed 20.11.13].
[33] Component price list acquired from BUNDU power. Available from: www.bundupower.co.za; 2013 [accessed 20.11.13].
[34] Darrieus turbine. Alternative hydro solutions. Available from: http://www.althydrosolutions.com; 2013 [accessed 27.11.13].
[35] Dalton GJ, Lockington DA, Baldock TE. Case study feasibility analysis of renewable energy supply options for small to medium-sized tourist accommodations. Renewable Energy 2009;34(4):1134–44.